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Controlling Mass Transport in Direct Carbon Dioxide Zero-Gap Electrolyzers via Cell Compression

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Abstract

The development of high-performance CO_2 electrolyzers is crucial for accelerating sustainable production of fuels and chemicals integrated with renewable energy sources. Here, we introduce a methodology to actively control mass transport inside a realistic zero-gap membrane-electrode-assembly CO_2 electrolyzer by varying the gasket thickness, which consequently changes cell compression. This allows control over the thickness and porosity of the gas diffusion electrodes, influencing the overall electrolyzer performance, as demonstrated using Ag-deposited electrodes. At low operating voltages (< 2.9V), both high and low compression electrolyzers exhibit similar faradaic efficiencies and partial current densities for CO formation. However, at high voltages, the low compression electrolyzer with high electrode porosity demonstrates superior CO selectivity and activity with suppressed H_2 formation. These experimental results are validated by the computational MEA model developed using the measured in-situ electrode thicknesses and electrode porosities. Additionally, liquid electrolyte saturation at the catalyst layer is found to play a dominant role in determining the mass transport, resulting in decreased electrolyzer performance with low electrode porosity. The systematic investigation in this study improves the understanding of the transport dynamics in MEA-based devices and provides insights into optimizing device design parameters for industry relevant CO_2 electrolysis.

Keywords

Carbon dioxide reduction, Gas diffusion electrode, Membrane electrode assembly, Mass transport, Porosity, Electrochemistry, Computational modeling, Carbon conversion

Introduction

The field of electrochemical CO_2 conversion has made substantial research efforts to address present sustainability and environmental challenges, introducing methods for producing fuels and chemicals at industrially relevant scales through the utilization of renewable energy sources like solar, wind, and hydropower.[1–3] The sustainable production of value-added products such as CO via the electrolysis of CO_2 can significantly help reduce reliance on fossil fuels and offer a sustainable alternative to conventional manufacturing processes, such as steam methane reforming and coal gasification, which require high temperatures (> 900 K) and high pressures (> 1.5 MPa).[4] Moreover, cost projections based on current CO_2 electrolysis technology indicate only a 17% higher cost than the current market price for CO ($0.15/kg),[5] and it is likely to become even more cost-competitive as the sustainable electrolysis technology continues to develop.

The overall performance of electrochemical CO_2 reduction is heavily influenced by the combination of complex reaction dynamics and interface processes.[6] Of particular interest is mass transport phenomena, which directly affect the activity and selectivity of CO_2 reduction and the competing H_2 evolution reaction.[7–9] Using gas diffusion electrode (GDE) structures enables gas-phase CO_2 feed, resulting in significantly improved mass transport and thus high current densities towards the formation of single- and multi-carbon products.[10–13] While GDE integrations help accelerate demonstration of high electrolyzer performance, much remains to be understood about transport phenomena and its impact on the catalytic reactions.

To better control and understand CO_2 mass transport, this work leverages the zero-gap membrane electrode assembly (MEA) electrochemical architecture, which consists of two electrodes on either side of the membrane that are compressed inside a cell housing.[14–16] The advantages of using zero-gap MEAs have recently been demonstrated for CO_2 reduction with improved product formation rates,

CO_2 utilization, and cell voltages.[16–19] This study, in particular, introduces a methodology to directly control the CO_2 mass transport by varying the applied cell compression to the MEA through the modification of the gasket thickness, inspired by device optimizations performed for hydrogen fuel cells, water electrolyzers, and redox flow batteries.[20–22] The changes in CO_2 mass transport ultimately dictate the voltage and current density at which the electrolyzer operates, which provides insights into operating conditions that lead to high energy efficiencies for more sustainable energy use. Specifically, controlling the mass transport in the GDEs coated with Ag catalyst leads to diverging CO_2 to CO conversion activity and selectivity trends, which are explained based on the characterization of the electrode thickness and porosity, and the computational modeling of the MEA. The findings in this work provide valuable insights into the electrode structure-performance relationships in realistic CO_2 electrolyzers, and more generally to large-scale zero-gap MEA-based electrochemical energy and chemical conversion devices.

Results and Discussion

The methodology developed in this study allows for actively controlling mass transport in the direct CO_2 zero-gap MEA electrolyzer with the utilization of varying gasket thicknesses (**Figure 1a**). Thick gaskets lead to lower cell compression and higher MEA thickness (**Figure 1b**), while thin gaskets lead to higher cell compression and lower MEA thickness (**Figure 1c**). The final MEA thickness is determined by the total thickness of the gaskets, which are made of polytetrafluoroethylene (PTFE) that is nearly incompressible, and the membrane, which separates the cathode and anode. Depending on the MEA thickness and the porous structure of the GDEs that changes along with it, the mass transport of reactants and products can be tuned to control electrolyzer performance. To simulate the changing electrolyzer selectivity towards the production of CO using Ag-deposited cathode GDE, a computational model with domains that represent each component in the zero-gap MEA is developed (**Figure S1**), showing predicted CO faradaic efficiency (FE) values across a realistic operating current density (**Figure 1d**). The CO FEs demonstrate nearly 90% at the low current densities then decrease with increasing current density, as

expected from Ag-based systems. Increasing the MEA compression leads to an earlier on-set current density for the CO FE loss, but the rate decrease remains to be similar. The model is also used to predict the saturation of electrolyte in the catalyst layer (CL), which shows a higher degree of wetting and a steeper increase for MEAs with higher compression with increasing current density (**Figure 1e**). The computational simulation of the compression-induced changes (i.e., thickness, porosity, etc.) provides insights into the changing local reaction environment and how this ultimately affects the electrolyzer performance in terms of product selectivity and current density.

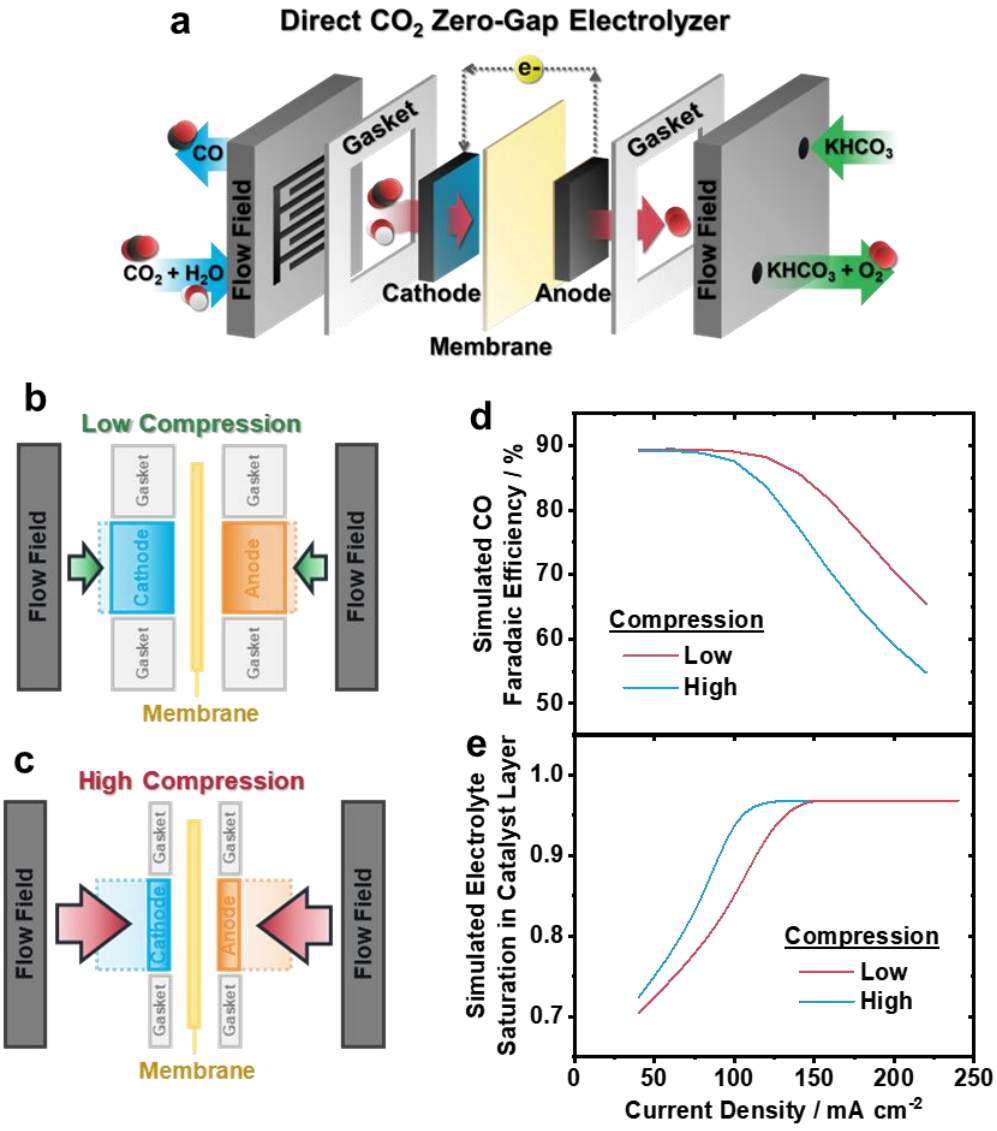


Figure 1. a) The schematic of the direct CO_2 zero-gap MEA electrolyzer used to control mass transport by varying gasket thickness. Cross-sectional schematics of the electrolyzer with b) thick and c) thin gaskets, leading to low and high MEA compressions, respectively (dashed and solid lines indicate thicknesses before and after compression, respectively). One-dimensional steady-state computational modeling of the electrolyzer demonstrating simulated d) CO FE, and e) electrolyte saturation in the CL under varying MEA compressions.

The compression that an MEA inside an electrolyzer experiences is directly measured by integrating a pressure sensitive film into the assembly which changes its color intensity upon applied compression (**Figure 2a**). The analysis of the pressure films compressed with gasket thickness increasing from 0.008 to 0.011" results in in-situ measured compression that decreases from 396 to 72 PSI (**Figure 2b**). The rate of decrease in the in-situ compression is moderate between relatively thin gaskets (i.e. -33 and -17%), while it is more drastic (-64%) between the thicker gaskets. This likely indicates a non-linear compression of the porous network in the GDEs, but the general trend of decreasing MEA in-situ compression with increasing gasket thickness is confirmed.

Thickness is another important change that is measured in-situ for each of the components in the MEA (i.e. cathode, anode, and membrane) using a compression tester (**Figure S2**). The resulting force-displacement curves obtained for each MEA shows two distinct slopes (**Figure 2c**). The shallower slope at the low force range is primarily attributed to the compression of the cathode and anode GDEs, which require relatively less force to compress due to the high porosity. The steeper slope at the high force range is due to the compression of the nearly incompressible PTFE gaskets and the membrane, resulting in only a slight thickness displacement. Thus, the force at which the slope changes marks the end of the MEA compression, and this force value is used to determine the in-situ thickness of the individual components based on its respective force-displacement curve (**Figure S3 and Table S1**). Specifically, the in-situ cathode GDE thickness is found to increase with increasing gasket thickness (**Figures 2d**) with smaller changes

between thinner gaskets (12% and 9.4%) followed a larger change with the thicker gaskets (15%), consistent with the trend observed with the in-situ compression observed above. The GDE compression is corroborated by the cross-sectional SEM images, which shows a visibly thinner carbon fiber layer (CFL) due to its high porosity and low bulk density, while less porous and more compact microporous layer (MPL) remains unchanged under compression (**Figure S4**).

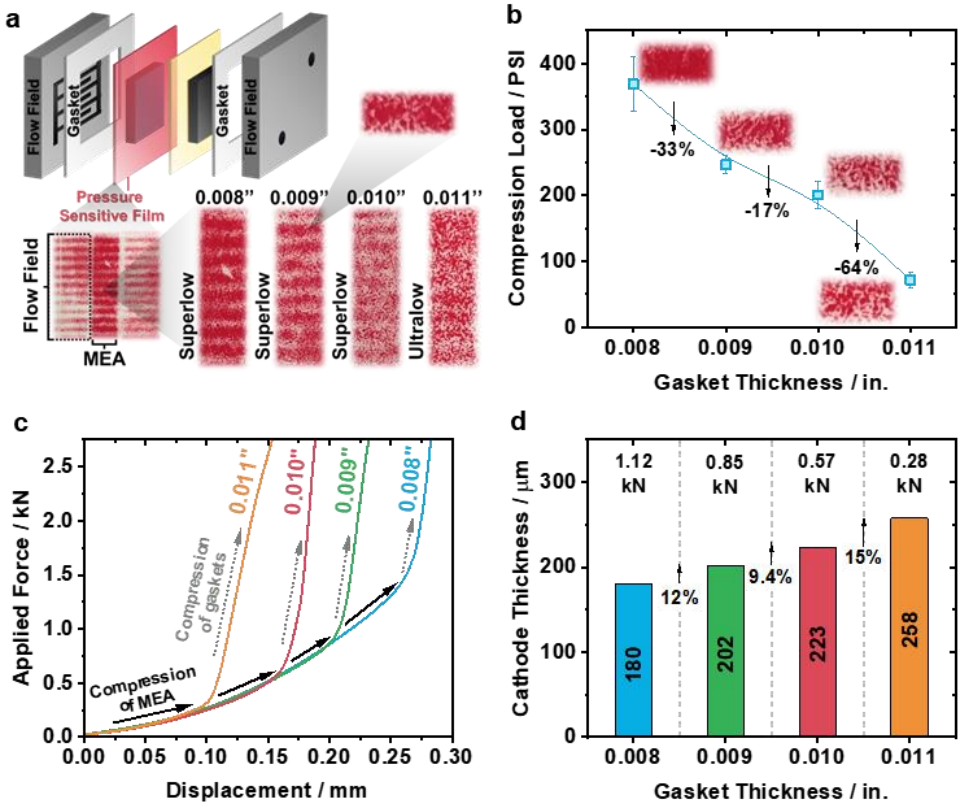
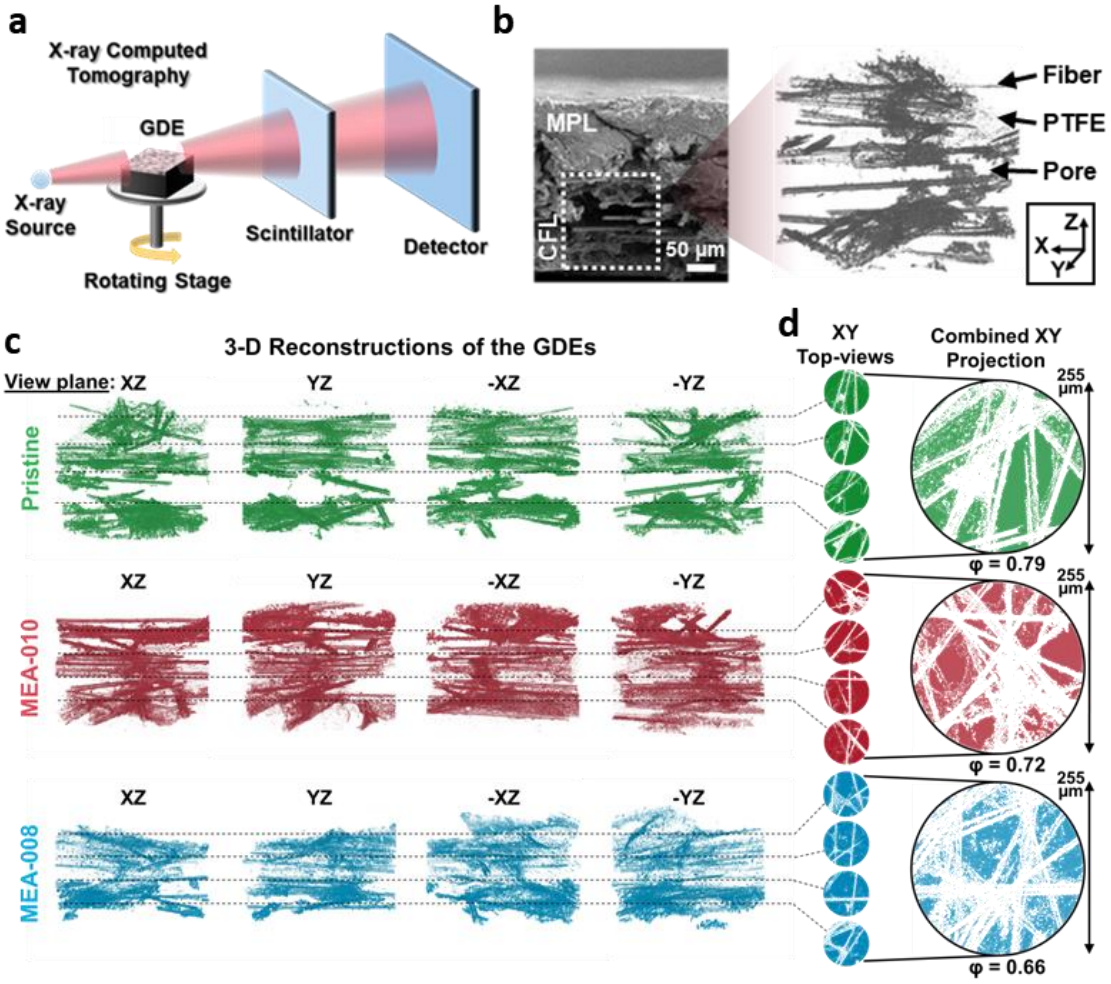


Figure 2. a) The schematic of the in-situ MEA compression measurement setup, with optical images showing the color intensity change of the pressure sensitive films recovered after performing compression using 0.008, 0.009, 0.010, and 0.011" thick gaskets. b) Measured compression as a function of gasket thickness. c) Force-displacement curves of the MEAs obtained using compression tester. d) Calculated in-situ cathode thicknesses as a function of gasket thickness.

More detailed structural study of the cathode GDEs is performed by leveraging X-ray computed tomography (XCT) (**Figure 3a**), which allows for 3-dimensional characterization of the carbon fibers and PTFE binders, which are difficult to visualize under conventional SEM analysis (**Figure 3b**). The 3-dimensional reconstructions of a pristine (uncompressed and not tested) GDE, and the GDEs recovered after electrolyzer testing with 0.008" (MEA-008) and 0.010" (MEA-010) gaskets, are visualized from four view planes (**Figure 3c**), showing decreasing thickness with increasing compression, consistent with the above in-situ measurements. Additionally, the electrodes show closer fiber-to-fiber distance and more aligned fibers, resulting in 'flattening' of the pores, which highlights the complex GDE structure and how structural changes can affect mass transport. In this study, electrode porosity is studied as the primary descriptor, which is defined as the volume ratio of pores-to-materials, which includes carbon fibers and PTFE binders. The average porosities are calculated by combining four equally spaced XY-plane images obtained along the Z-axis for each GDE, resulting in 0.79, 0.72, and 0.66 for pristine, MEA-010, and MEA-008 GDEs, respectively (**Figure 3d**). While these values are obtained ex-situ and fluctuations are observed in porosity of individual XY-planes along the electrode depth (**Figure S5**), the quantitative analysis of the porous structure allows for better understanding of the mass transport effects on CO_2 electrolyzer performance.



145

146 **Figure 3.** a) The illustration of the XCT setup for 3-dimensional cathode GDE characterization. b) Cross-
147 sectional SEM of a pristine GDE (left), and the corresponding 3-dimensional reconstruction of the CFL
148 obtained using XCT (right). c) 3-dimensional reconstructions of the pristine GDE (top, green), and the GDEs
149 recovered after electrolyzer testing using 0.010" (middle, red) and 0.008" (bottom, blue) gaskets
150 visualized from four view planes (XZ, YZ, -XZ, and -YZ, where Z is the axis that is perpendicular to the GDE
151 surface). d) four XY top-view images along the electrode depth (Z-axis), and the combined top-view images
152 with the corresponding calculated electrode porosities.

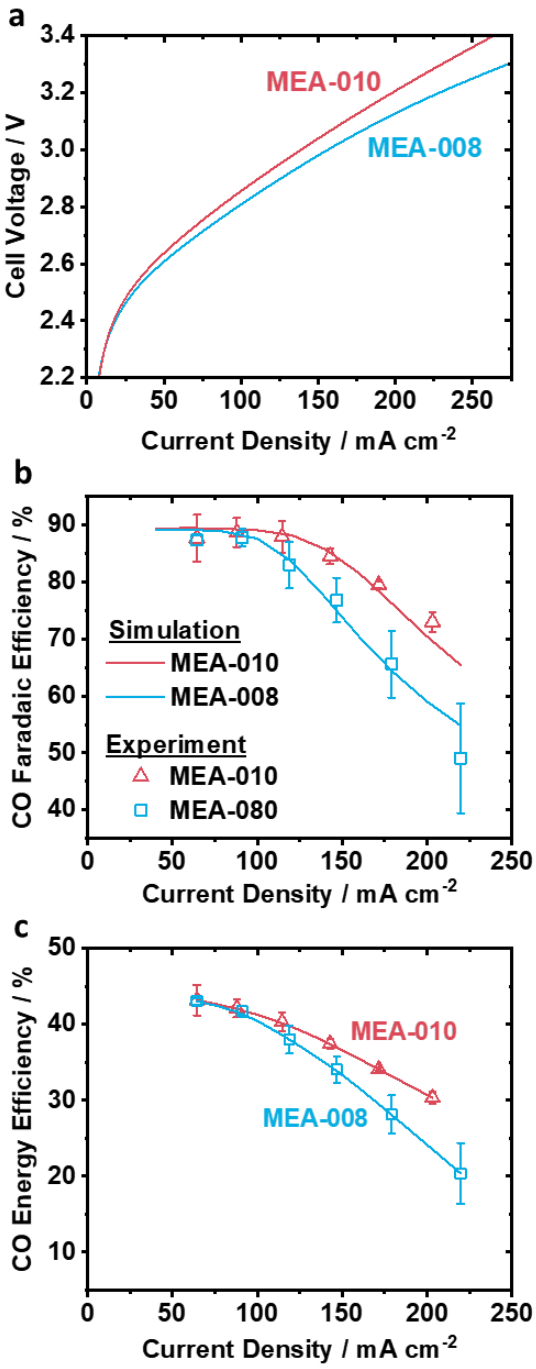
153 Based on the high-frequency resistance measurement of the MEAs, there is a monotonic increase
154 in cell resistance from 0.98 to 1.2 Ω cm^2 with increasing gasket thickness (**Figure S6**). Specifically, the more

compressed MEA-008 shows 22% lower resistance compared to the less compressed MEA-010 due to the decreased packing density of conductive carbon fibers in the GDE (**Figure S7**).[23] Consequently, MEA-008 operates under more favorable voltage than MEA-010, demonstrating up to 101 mV improvement at 240 mA cm^{-2} (**Figure 4a**). Based on the voltage-dependent CO FE plot (**Figure S8a**), both MEAs perform similarly high CO FEs in the range 87-89%, consistent with other reported Ag-based systems,[24–27] at the low operating voltages. However, at high voltages, MEA-010 better maintains CO FEs than MEA-008 (73 vs 49% at 3.2 V, respectively), resulting in the divergence of CO selectivity trends. Similarly, the CO partial current density (PCD) curves perform nearly identical at low voltages but diverge with increasing voltage with MEA-010 demonstrating higher current density of 148 $mAcm^{-2}$ compared to 103 $mAcm^{-2}$ at 3.2 V obtained with MEA-008 (**Figure S8b**). While MEA-010 demonstrates continuously increasing CO PCDs, MEA-008 shows a peak at 3.1 V and a loss at 3.2 V likely due to the mass transport limited operation. The greater CO selectivity and activity observed with MEA-010 leads to generally lower H_2 FE (**Figure S9a**) and PCD (**Figure S9b**) particularly at high voltages compared to MEA-008. iR-correcting the voltages eliminates the effects of changing cell resistance and further highlights the diverging selectivity and activity trends observed between the two MEAs, suggesting compression-induced mass transport as a main contributor that can modulate MEA performance (**Figure S10**). At the electrode level, this effectively translates to intrinsic porosity differences that allows the investigation of GDE structures for overcoming the adverse effects of larger pH gradient and CO_2 losses through OH^- neutralization that occur in CO_2 electrolyzers.[28]

The experimental CO FE results are shown to closely follow the simulated selectivity trends observed for both MEA-010 and MEA-008 (**Figure 4b**), based on the computational model developed using the measured average electrode porosities and thicknesses (**Table S2**), along with the reported kinetic parameters, operating conditions, and material properties (**Table S3 and S4**). The electrolyte saturation profile of the MPL is obtained using the previously reported polymer electrolyte fuel cell MEA data,[29,30] which shifts higher with decreasing electrode porosity. This agrees with more rapidly increasing H_2

formation observed with MEA-008, which likely stems from the flooding of the active interface, which is corroborated by a larger double layer capacitance observed compared to MEA-010 (**Figure S11**). This finding is important as flooding is known to play a major role in limiting the CO_2 access for the catalyst and accelerating HER, in addition to other reported degradation mechanisms, such as changes in the local pH and the catalyst structure and morphology.[31,32] Furthermore, the higher energy efficiencies observed for CO formation with MEA-010 compared to MEA-008 indicate a more sustainable operation of the electrolyzer achieved through improved CO_2 transport by leveraging control over cell compression (**Figure 4c**). The continuous operation of MEA-010 at the electrolyzer voltage that yields the highest CO selectivity results in CO FE and electrolyzer current density losses over a span of 12 hours (**Figure S12**). This is expected as the PVD Ag-deposited GDEs employed in this study are not optimized for long-term electrolyzer stability, despite their high uniformity and reproducibility, which allows for the investigation of mass transport effects. However, this finding serves as motivation for future research aimed at optimizing electrode structure and composition to enhance long-term electrolyzer stability.

Despite the likely increase in the porosity of MEA-011 with lowered cell compression, the resulting CO FEs are lower compared to those of MEA-010 (**Figure S13**). This is attributed to a 15% increase in the electrode thickness for MEA-011 (258 um) compared to MEA-010 (223 um) (**Figure 2d**), which leads to a significantly longer diffusion length for CO_2 to reach the active interface. This highlights the tradeoff that exists between the electrode thickness and porosity, both of which are affected by cell compression, but have opposing effects on the performance of CO_2 electrolyzer.



198

199 **Figure 4.** a) Polarization curves obtained from testing MEA-010 and MEA-008 with an average porosity of
200 0.72 and 0.66, respectively. b) Experimentally obtained CO selectivity profiles validated by the model
201 simulation results. c) Energy efficiencies towards the formation of CO. The error bars represent the
202 standard deviation from testing three or more independent MEAs.

Comparing porosity-dependent performance at the mass transport limited regime where the electrolyzer operates at high current densities and voltages, the increase in electrode porosity is found to help improve CO FE and PCD, while suppressing H_2 FE and PCD (**Figure 5a and 5b**). The benefits of increased porosity are uniquely observed at the mass transport limited regime since the activity and selectivity towards CO and H_2 are found to be very similar at low to moderate operating current densities and voltages (<100 mA cm^{-2} total current density, <2.8 V) (**Figure S14a and S14b**). These observations indicate that the electrolyzer performance is more governed by micro-environmental effects such as local pH and CO_2 concentrations at low voltages,[33,34] while macro-scale properties like electrode porosity leads to varying mass transport limitations influencing the overall electrolyzer performance.

Despite the above results indicating higher porosity electrodes better facilitating CO_2 mass transport and the HER suppression, some intermediate compression is still necessary to maintain a gas-tight seal with the lowest possible cell resistance. In addition to the reduced effective CO_2 gas diffusivity and gas permeability with decreasing porosity, mass transport limitations are dominated by the presence of liquid electrolyte in the catalyst layer as predicted by the model (**Figure 1e**). As shown schematically, the high electrode porosity benefits from relatively lower electrolyte saturation at the catalyst layer (**Figure 5c**), resulting in a shorter diffusion length for improved CO formation and suppressed HER. In contrast, the low electrode porosity is negatively impacted by higher saturation, which likely leads to electrolyte penetration further into the catalyst layer (**Figure 5d**), resulting in decreased CO selectivity and activity coupled with accelerated HER, consistent with the reported failure modes via flooding and carbonate salt formation.[35,36]

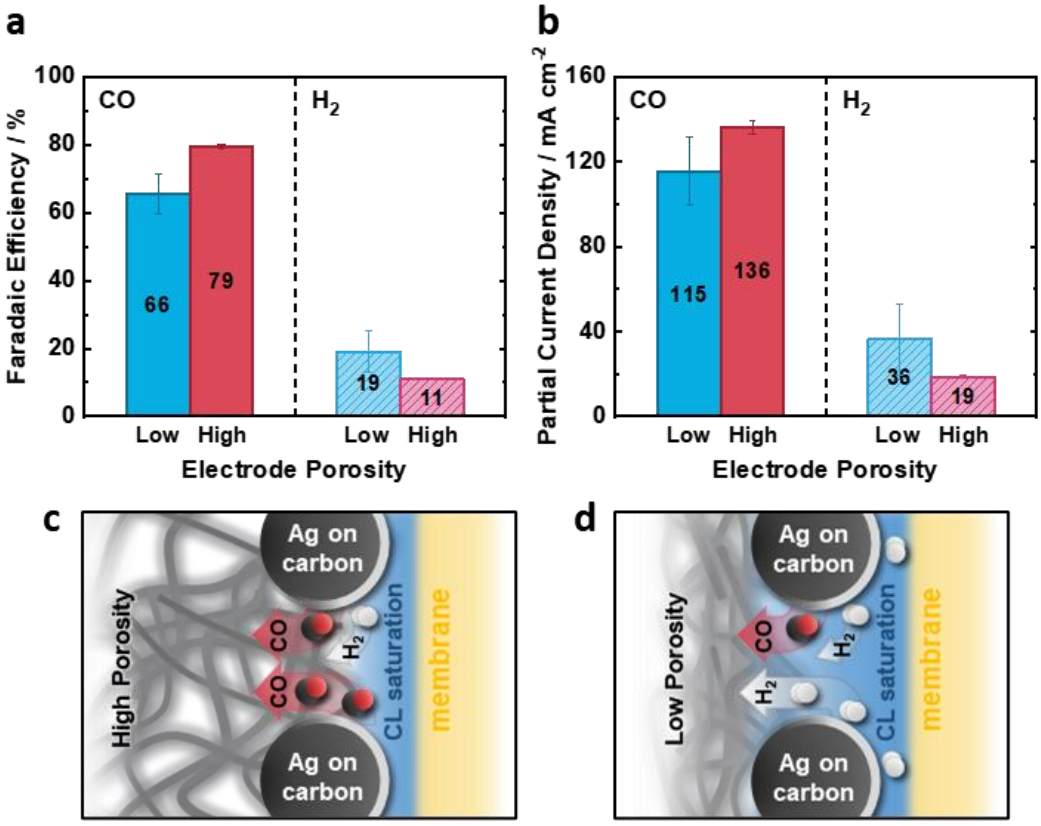


Figure 5. Comparison of the CO and H_2 a) FEs and b) PCDs obtained with 0.66 (low) and 0.72 (high) porosity GDEs at the mass transport limited operating regime (>170 mA cm^{-2} total current density, >3.1 V). The schematic illustrations of the changes in the saturation of the catalyst layer at the electrode interface at c) low and d) high MEA compression, leading to varying selectivity and activity towards CO and H_2 formation.

Conclusions

In summary, our study demonstrates the critical role of controlling mass transport via cell compression in industrially relevant direct CO_2 to CO zero-gap MEA electrolyzers. The mass transport control is performed through systematic variation in gasket thickness, resulting in GDE thicknesses ranging from 330 (uncompressed) to 180 µm (most compressed). Furthermore, the quantitative XCT analysis enables measurement of electrode porosity, resulting in 0.66 and 0.79 for high and low cell compressions,

respectively. Electrochemical evaluation reveals that while both electrolyzers exhibit similar faradaic efficiencies and partial current densities towards CO production at low operating voltages (<2.9 V), the low compression electrolyzer demonstrates significantly higher CO selectivity and activity at high operating voltages. The computational model of the MEA developed using the measured electrode properties not only validates the experimental electrolyzer performance, but also suggests reduced effective CO_2 gas diffusivity and gas permeability with decreasing electrode porosity. Furthermore, the model predicts increasing electrolyte saturation at the catalyst layer with increasing porosity, limiting the mass transport of CO_2 which results in decreased CO activity and selectivity coupled with accelerated HER. This work highlights the significance of cell compression as a critical design parameter for understanding mass transport in zero-gap MEA-based devices and provides valuable insights into further improving device and electrode designs for high-performance energy and chemical conversion systems.

Supporting Information

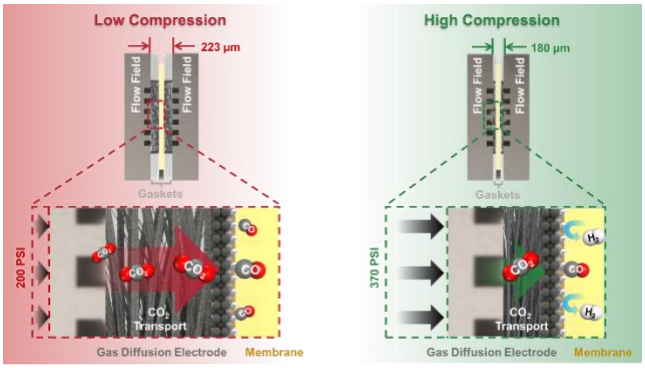
Detailed descriptions of experimental and computational methods, computational modeling domain for a zero-gap MEA CO_2 electrolyzer (Figure S1), in-situ thickness measurement setup (Figure S2), the force-displacement curves (Figure S3), Cross-sectional SEM images (Figure S4), XCT porosity measurements (Figure S5), High frequency resistance measurements (Figure S6-S7), voltage-dependent electrolyzer performance (Figure S8-S10), Double-layer capacitance (Figure S11), continuous MEA operation (Figure S12), MEA-010 and MEA-011 CO selectivity comparison (Figure S13), porosity-dependent performance (Figure S14), MEA and component thicknesses (Table S1), measured porosity and in-situ thicknesses (Table S2), kinetic parameters (Table S3), and operating conditions and material properties (Table S4) used for computational modeling.

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Table of Contents



Synopsis

The effects of mass transport on the realistic carbon dioxide electrolyzer device performance are investigated to accelerate sustainable fuels and chemicals production.

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